Multimodal Skin Lesion Classification and Screening with an Improved EfficientNetV2 Ensemble

ABSTRACT

This study presents an improved EfficientNetV2-S ensemble for eleven-class skin lesion classification and broader lesion screening. Clinical photographs, dermoscopic images and patient metadata supply complementary inputs. The trained image specialists and metadata multilayer perceptron are retained. A class-balanced classifier over compressed visual features, metadata and specialist scores is combined with six-view image prediction averaging through a fixed geometric mean.

On the same 786 MILK10k validation lesions, the ensemble increases exact-class accuracy from 59.67% to 69.21% and macro F1 from 0.3551 to 0.4523. It corrects 103 previous errors while regressing 28 correct decisions. Broader screening groups the leading diagnosis into malignant/precancerous or benign/inflammatory families. Under this single presentation rule, accuracy rises from 80.41% to 82.70% and specificity from 51.79% to 66.96%, while sensitivity decreases from 91.81% to 88.97%. These results describe a measured aggregate gain with diagnosis-specific trade-offs.

Exploratory data analysis, class-level confusion analysis and numerical replay establish the evidence supporting the comparison. A Streamlit demonstration integrates upload relevance checking, modality comparisons, inconclusive outputs and specialist heatmaps. The saved original baseline and historical JI-ADF adaptation results remain separate and unchanged. The principal contribution is the evaluated multimodal ensemble and interpretation pipeline. The validation study remains exploratory because of earlier checkpoint selection and sequential development; further evaluation should use independently grouped data and a locked selection procedure.

Keywords: multimodal learning; MILK10k; EfficientNetV2; frozen features; test-time augmentation; broader lesion screening.

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Abbreviations and Key Terms

Term	Meaning
JI-ADF	Joint-Individual Learning with Adaptive Decision Fusion; the methodological reference.
MLP	Multilayer perceptron, used for patient metadata.
PCA	Principal-component analysis, fitted on training image features.
TTA	Test-time augmentation: averaging predictions across image transformations.
OOF	Out-of-fold predictions from rows excluded from a fold's downstream fit.
CLAHE / ROI	Contrast-limited adaptive histogram equalization / region of interest.
Macro F1	Mean of class-specific F1 scores, giving every class equal weight.
Sensitivity	Fraction of positive-reference lesions flagged positive.
Specificity	Fraction of negative-reference lesions classified negative.
ROC-AUC	Area under the receiver-operating-characteristic curve.
Broader screening	Malignant/precancerous-family versus benign/inflammatory-family grouping, rather than confirmed malignancy.
Grad-CAM	Gradient-based spatial explanation of an individual image classifier.

CHAPTER 1

INTRODUCTION

Clinical photographs, dermoscopy and patient metadata provide complementary evidence for skin lesion assessment. MILK10k pairs these inputs across malignant diagnoses, benign conditions and simulators [1]. This study develops an improved EfficientNetV2-S ensemble combining frozen image representations, metadata and transformed-image predictions. Class imbalance motivates reporting macro F1, class recall and screening errors alongside overall accuracy.

1.1 Problem and Motivation

The primary task predicts one of 11 diagnostic categories from a paired lesion record. A secondary task groups those outputs into malignant/precancerous and benign/inflammatory families. A wrong diagnosis can still produce the correct broad family, so the endpoints are evaluated separately. An independent relevance validator screens unsupported uploads before disease inference; acceptance does not verify the predicted diagnosis.

1.2 Objectives and Contribution

The objective is to improve the saved modular EfficientNetV2 pipeline while preserving its trained specialists. Contributions are the evaluated feature/TTA ensemble, matched-cohort comparisons, class and screening error analyses, and an integrated interpretation interface. JI-ADF supplies a related architectural reference; its local adaptation remains historical context. The contribution concerns the evaluated integration of established methods rather than a new convolutional backbone.

1.3 Scope

The main experiment uses one MILK10k release and fixed lesion partitions. Trained specialists, baseline checkpoints and historical results remain unchanged. Validation comparisons remain exploratory because the cohort guided earlier checkpoint selection and sequential development. The local JI-ADF adaptation is distinguished from the published method and its test scores.

On matched validation lesions, the extension raises macro F1 from 0.3551 to 0.4523 and exact-class accuracy from 59.67% to 69.21%. Grouping its leading diagnosis yields 82.70% broader screening accuracy. The interface follows this single policy; the archived summed-score rule is evaluated separately.

A separate direct-binary study appears in Appendix A. Its endpoint, cohorts and thresholds differ from the main evaluation, so its results are reported independently.

CHAPTER 2

RELATED WORKS

EfficientNetV2 supplies the image-backbone family for the main ensemble [2]. The proposed extension combines fixed visual representations and test-time prediction averaging with patient metadata. JI-ADF provides a multimodal architectural reference [3]; the measured historical comparator is our local adaptation. Shared backbones do not control differences in preprocessing, balancing, optimization or label use.

Multimodal architectural reference

In JI-ADF, paired images supply visual features and structured metadata supply contextual features. A joint stream models their interaction, while visual and metadata branches receive auxiliary supervision. A gating network derives sample-dependent weights from the branch logits and combines their class predictions. The MMFA module provides attention-based interaction and a residual path. This separates the mechanism for learning representations from the mechanism for combining decisions.

Our adaptation exposes clinical photographs, dermoscopy and metadata to this design. The original modular implementation instead trains image specialists independently and learns a later classifier over their predictions and the metadata prediction. These are different complete learning pipelines, even when they use the same backbone family. The later frozen-feature extension preserves the modular specialists and changes the downstream representation and aggregation stages.

Table 2.1. Research context and role in this implementation.

Method	Role
MILK10k	Paired images, metadata and 11 diagnostic categories.
JI-ADF	Reference multimodal method; separately trained local adaptation.
EfficientNetV2-S	Independent clinical and dermoscopic specialists.
PCA and logistic regression	Compact frozen features and a regularized downstream classifier.
TTA and Grad-CAM	Prediction averaging and individual image-model explanations.

Backbones and specialist regularization

EfficientNetV2 provides an efficient convolutional representation; it does not, by itself, determine how metadata enter the decision or how rare diagnoses are learned. In this project, two EfficientNetV2-S instances supply separate clinical and dermoscopic representations. Their specialization, preprocessing and saved weights are part of the experimental configuration, rather than interchangeable components.

Mixup, label smoothing and AdamW regularize the original specialist training [4–6]. This report treats their combination as an implemented recipe rather than isolating a causal contribution for each component. The independent ResNet18 relevance gate follows the residual-network architecture [7]; its acceptance and rejection results are evaluated separately from classification among the eleven diagnoses.

Frozen representations and downstream fusion

An image classifier's eleven scores summarize the evidence required by its existing output head. A higher-dimensional frozen embedding can retain information that the head does not express in those scores. The extension therefore evaluates a classifier on compressed image features, encoded metadata and specialist scores. This changes what reaches the downstream learner while preserving the original image encoders.

PCA compresses representations without using diagnosis labels [8]. Its role here is dimensionality reduction, not supervised selection of disease features. A scaler and regularized logistic classifier subsequently learn from the permitted fitting lesions. The distinction matters: frozen neural weights do not imply that the complete extension is untrained, because the downstream classifier is newly fitted.

Prediction averaging and complementary evidence

Test-time augmentation aggregates predictions across transformed inputs [9]. This implementation uses a fixed six-view average rather than a learned augmentation policy. The feature classifier receives identity-view embeddings, whereas the image-averaging member combines predictions across views. Their geometric combination can therefore draw on different summaries of the same photographs without adding another image backbone.

Table 2.2. Interpretation of the three downstream strategies evaluated in this project.

Strategy	Evidence available	Interpretive distinction
Score-level fusion	Three eleven-class specialist outputs	Reweights existing predictions; does not recover discarded embedding information.
Frozen-feature classifier	Compressed image embeddings, metadata and specialist scores	Fits a new decision boundary; uses more downstream fitting labels than the original score stack.
Feature/TTA ensemble	Feature-classifier output and averaged image predictions	Combines two fixed members; does not learn a new visual representation.

Explanation and the scope of comparison

Grad-CAM localizes spatial influence on an image branch's target logit [10]. It does not explain the complete PCA/logistic ensemble, establish a lesion boundary or verify that a diagnosis is correct. The demonstration accordingly labels each heatmap with the image specialist and category it explains. The model-family comparison and explanation interface are complementary forms of analysis, rather than independent evidence of diagnostic accuracy.

The comparisons reported below concern this project's recorded configurations and fixed cohorts. They support claims about measured pipeline behavior. They do not establish that an independently implemented adaptation reproduces the published method's benchmark performance, or that a simpler fusion equation alone caused the observed gain. This framing allows the JI-ADF reference, original implementation and frozen extension to be discussed within one research narrative without conflating their evidence.

CHAPTER 3

PROPOSED WORK

3.1 Dataset and Exploratory Analysis

The release contains 10,480 images from 5,240 lesions, with one clinical close-up and one dermoscopic image per lesion. Targets and shared metadata are consistent within every pair; no lesion identifier crosses partitions. Patient identifiers are unavailable, so this does not establish patient-level separation. Every distributed image has a readable 600-by-450-pixel header. Header readability is a file-integrity observation, not a semantic image-quality assessment.

Table 3.1. Canonical diagnosis order and lesion counts in the selected release.

Code	Diagnostic family	Lesions
AKIEC	Actinic keratosis / intraepidermal carcinoma	303
BCC	Basal cell carcinoma	2,522
BEN_OTH	Other benign proliferations	44
BKL	Benign keratinocytic lesion	544
DF	Dermatofibroma	52
INF	Inflammatory and infectious conditions	50
MAL_OTH	Other malignant proliferations	9
MEL	Melanoma	450
NV	Melanocytic nevus	746
SCCKA	Squamous cell carcinoma / keratoacanthoma	473
VASC	Vascular lesions and hemorrhage	47

BCC contributes 48.13% of lesions; its support is 280.2 times that of MAL_OTH. Five categories contain fewer than 100 lesions. The split matrix shows how this imbalance persists after partitioning: the validation set contains 378 BCC lesions but one MAL_OTH lesion. All evaluations retain the complete canonical class order.

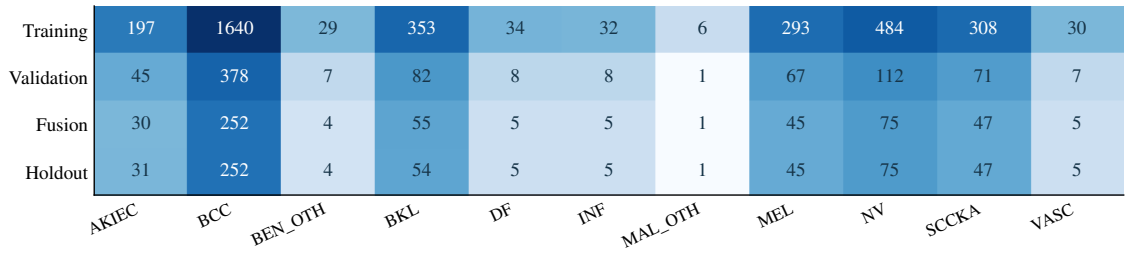


Figure 3.1. Lesion counts by diagnosis and partition. Colours use a logarithmic scale; annotations are actual counts.

Metadata are summarized once per lesion. Age is missing for 20 lesions (0.38%) and site for 31 (0.59%); sex and skin tone have no null values. Median recorded age is 65 years, with interquartile range 50–75. Skin-tone category 3 accounts for 60.57% of lesions, while category 0 has six. These release-specific 0–5 categories are not Fitzpatrick types. The distributions describe the available cohort; they do not demonstrate equivalent model performance across groups.

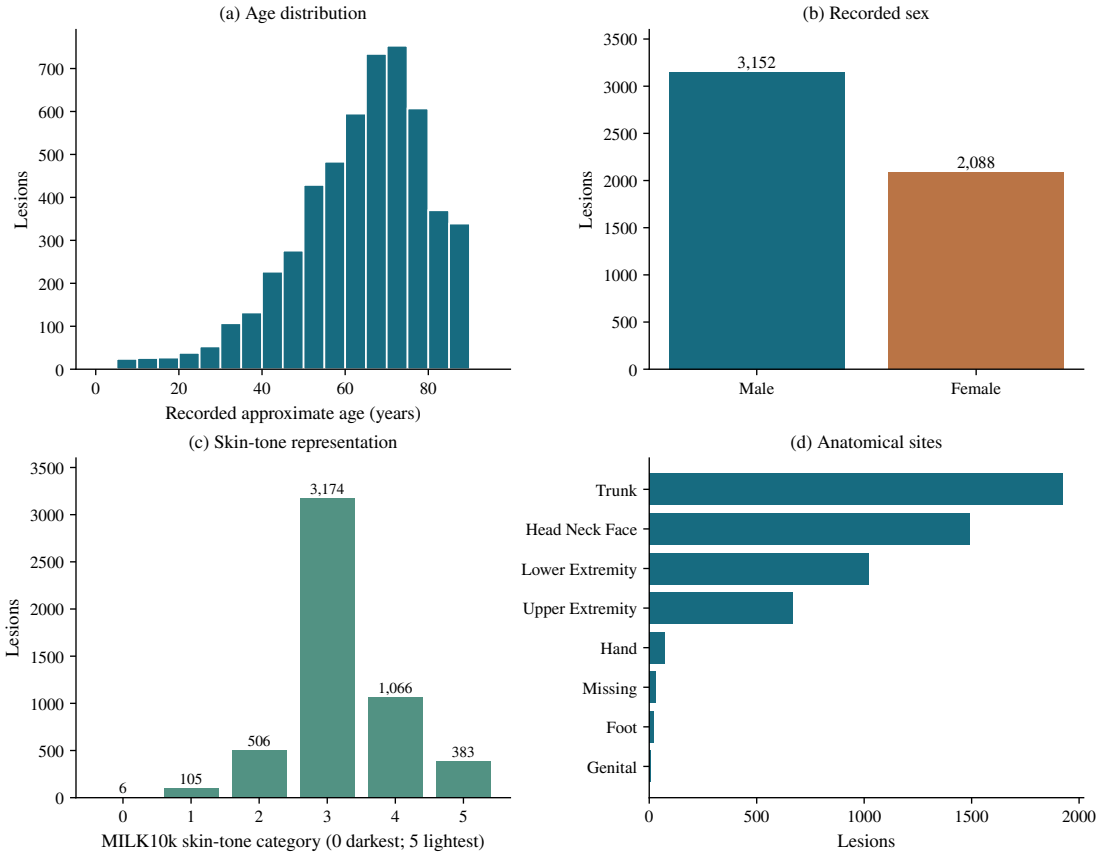


Figure 3.2. Lesion-level age, sex, skin-tone and anatomical-site distributions. Missing site is explicit; missing ages are omitted from the histogram.

3.2 Preprocessing and Specialist Training

The metadata encoder uses age, skin tone, sex and site. Numeric imputation, scaling and categorical vocabularies are fitted on training lesions only. Diagnosis, pathology confirmation and acquisition attribution are excluded. The saved encoding has 12 dimensions. Unknown upload fields remain missing and pass through this same encoder rather than being filled with a reference diagnosis or a manually chosen class-associated value.

Both image branches use luminance CLAHE, a heuristic foreground crop, smoothing, 224-pixel resizing and ImageNet normalization. Dermoscopy additionally uses hair-mask estimation and inpainting. The foreground operation is a heuristic crop, not trained segmentation. Training uses flips and small rotations; evaluation preparation is deterministic. Interactive uploads pass the relevance gate before disease processing; rejected inputs produce neither screening nor diagnosis scores.

The original specialists use inverse-square-root frequency sampling and loss weights, classifier-head warm-up, dropout, Mixup and label smoothing. Each EfficientNetV2-S supplies 1,280 pooled features and 11 logits. The metadata MLP has a 128-unit hidden layer. Original training completes 27 clinical, 18 dermoscopic and nine metadata epochs. All these weights remain fixed during the extension.

3.3 Original Fusion and Frozen-Feature Extension

The proposed feature/TTA ensemble retains all three inputs and the saved specialist weights.

Original fusion is class-balanced logistic regression ($C = 0.05$) on 33 concatenated specialist scores, fitted on 524 fusion lesions. The extension concatenates 64 whitened PCA coordinates per image, the 12 metadata features and all 33 specialist scores:

$$\mathbf{h} = [\mathrm{PCA}_{c,64}(\mathbf{f}_c); \mathrm{PCA}_{d,64}(\mathbf{f}_d); \mathbf{m}; \mathbf{p}_c; \mathbf{p}_d; \mathbf{p}_m] \in \mathbb{R}^{173}. \tag{3.1}$$

PCA fits only the 3,406 training lesions. A downstream scaler and class-balanced logistic regression ($C = 0.1$) produce $\mathbf{q}^F$. The second member, $\mathbf{q}^T$, averages each image's predictions over identity, horizontal/vertical flips and 90°, 180°, 270° rotations, then averages the two modalities equally. Metadata enter through the feature member.

$$q_k = \frac{\sqrt{\max(q_k^F, \epsilon)\,\max(q_k^T, \epsilon)}}{\sum_{j=1}^{11} \sqrt{\max(q_j^F, \epsilon)\,\max(q_j^T, \epsilon)}}, \qquad \epsilon = 10^{-12}. \tag{3.2}$$

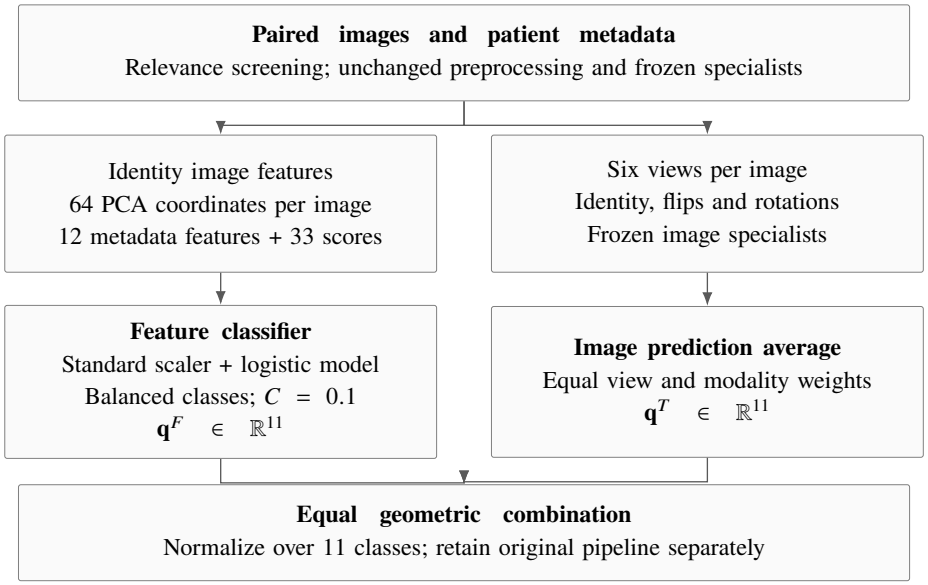


Figure 3.3. Feature/TTA ensemble with preserved clinical, dermoscopic and metadata specialists.

Equal geometric weights are fixed, without class thresholds or case routing. Paired inference uses twelve image forwards with identity features reused; heatmaps add passes.

3.4 Broader Screening Rules

The positive-family set is $\mathcal{P} = \{\text{AKIEC}, \text{BCC}, \text{MAL_OTH}, \text{MEL}, \text{SCCKA}\}$. The other six categories form the benign/inflammatory family. AKIEC mixes actinic keratosis and intraepidermal carcinoma; SCCKA includes keratoacanthoma. Accordingly, this endpoint is broader malignant/precancerous-family screening, not confirmed cancer versus benign disease.

Table 3.2. Declared screening families; all eleven diagnostic categories are retained.

Family	Included categories
Malignant / precancerous	AKIEC, BCC, MAL_OTH, MEL, SCCKA
Benign / inflammatory	BEN_OTH, BKL, DF, INF, NV, VASC

Two fixed rules are evaluated identically for both pipelines. The presentation policy preserves the existing eleven-class prediction and maps its category to a family. The archived alternative thresholds the sum of positive-family scores at 0.5:

$$s_+ = \sum_{k \in \mathcal{P}} q_k, \qquad \widehat{b}_{\text{sum}} = \mathbf{1}[s_+ \geq 0.5], \qquad \widehat{b}_{\text{top}} = \mathbf{1}[\arg\max_k q_k \in \mathcal{P}]. \tag{3.3}$$

Why the rules can disagree

Consider a hypothetical score vector assigning NV a score of 0.40 and three positive-family categories 0.20 each, with the remaining scores zero. Its leading diagnosis is NV, so category grouping returns the benign/inflammatory family. Summing the positive-family scores gives 0.60 and therefore a positive decision at 0.5. This example illustrates the rules; it is not a measured project result.

The summed score can be interpreted as accumulated model support for a family, but these outputs have not been calibrated as individual disease-risk probabilities. In particular, a conventional 0.5 cutoff need not provide a suitable operating point for each pipeline. The report therefore preserves both analyses while the demonstration consistently uses the leading-diagnosis rule. No cutoff was optimized on validation labels for this revision.

Evaluation consequences

Exact classification and family grouping use the same reference labels but different success criteria. A MEL lesion classified as BCC is an exact-diagnosis error and a correct positive-family decision. A MEL lesion classified as NV fails both endpoints. Conversely, a BKL lesion classified as NV is wrong at the eleven-class level but correct at the family level. These distinctions allow meaningful screening analysis without revising or replacing any original diagnostic prediction.

A supplementary analysis excludes AKIEC and SCCKA because of their mixed category definitions. Its scores are renormalized across the remaining categories and its cohort is reported separately. Excluding categories changes the question being evaluated; it is not used to replace the full-cohort result or to select the presentation policy after inspecting individual outcomes.

CHAPTER 4

EXPERIMENTAL SETUP AND RESULTS

4.1 Protocol and Evaluation

The fixed split contains 3,406 training, 786 validation, 524 fusion and 524 holdout lesions. Evaluation is lesion-level in the canonical eleven-class order. Original base-model checkpoints used validation macro F1. The principal comparison evaluates original score fusion and the improved feature/TTA ensemble on exactly the same 786 validation lesions. The separate 524-lesion historical comparison supplies context for the original saved implementation and JI-ADF adaptation.

Table 4.1. Evaluation units and permitted roles of the fixed lesion partitions.

Partition	Lesions	Role in the completed experiments
Training	3,406	Specialist training, metadata encoding and image PCA fitting.
Validation	786	Original checkpoint selection and later exploratory comparisons.
Fusion	524	Original score stacking and downstream cross-fitting/selection.
Holdout	524	Historical original/adaptation evaluation; later inspected examples.

What macro F1 measures

For category k, precision measures the proportion of its predictions that are correct, while recall measures the proportion of its reference lesions recovered. Macro F1 averages the class-specific harmonic means [11]:

$$F_{1,k} = \frac{2TP_k}{2TP_k + FP_k + FN_k}, \qquad F_{1,\text{macro}} = \frac{1}{11}\sum_{k=1}^{11} F_{1,k}. \tag{4.1}$$

The calculation retains all eleven categories and assigns zero to undefined class scores, matching the archived implementation. A macro F1 of 0.4523 is not an accuracy of 45.23%. Exact accuracy counts correct lesions; macro F1 gives each diagnostic category equal importance. This makes uncommon-category failures visible even when many common lesions are classified correctly.

Balanced accuracy averages class recall; in a binary task it is the mean of sensitivity and specificity [12]. Class support and the confusion matrix complement both aggregate metrics. For broader screening, sensitivity, specificity and predictive values are reported together because a change can reduce false alarms while increasing missed positive-family lesions.

Evidence and fitting boundaries

The frozen extension explored 92 alternatives across score fusion, features, neural heads and fixed ensembles. Final combination selection used three-fold fusion OOF predictions; component choices came from earlier exploratory rounds and were not nested within those folds. Feature classifiers refit on 3,930 train-plus-fusion lesions; original score stacking fits fusion rows only. Validation labels fit neither PCA nor the downstream classifier, but repeated validation inspection limits confirmatory inference [13].

4.2 Preserved 11-Class Results

The main result concerns the improved ensemble on the same validation cohort as its original score-fusion comparator [13]. All original image and metadata checkpoints remain fixed.

Table 4.2. Frozen-model alternatives on the same 786 validation lesions.

Method	Accuracy (%)	Balanced (%)	Macro F1
Saved original fusion	59.67	42.82	0.3551
Scaled score fusion	61.58	45.72	0.4040
Frozen feature LR	67.18	40.85	0.4151
Six-view image average	63.99	47.21	0.4017
Frozen feature MLP	69.72	39.34	0.4027
Feature/TTA ensemble	69.21	45.63	0.4523

On the same 786 validation lesions, the ensemble increases exact-class accuracy by 9.54 percentage points and macro F1 by 0.0972. It correctly classifies 544 lesions compared with 469 for original fusion. This is a meaningful measured improvement within the development cohort, while a macro F1 of 0.4523 still indicates uneven class-level behavior.

The frozen feature MLP achieves the highest validation accuracy, 69.72%, but its macro F1 is 0.4027 and balanced accuracy is 39.34%. The selected ensemble therefore has the strongest macro F1 in this retained comparison, rather than the highest value of every metric. Six-view averaging alone has balanced accuracy 47.21%, also exceeding the selected ensemble on that particular criterion.

Historical macro F1 of 0.4026 and ensemble macro F1 of 0.4523 describe different cohorts. The matched validation improvement is 0.3551 to 0.4523.

A separate replay of the saved ensemble scores gives macro Dice 0.4270 when each category is independently thresholded at 0.5, rather than using argmax. The paper reports Dice 0.505 on 479 official hidden test lesions [3]. This literature value is contextual: the different cohorts prevent a matched superiority claim. Neither value replaces the argmax results above.

Historical reference retained unchanged

The historical holdout remains unchanged [14, 15]. This 524-lesion comparison uses our independent JI-ADF adaptation, not the authors' checkpoint, and is separate from the ensemble validation study.

Table 4.3. Historical comparison on the same 524 holdout lesions, without TTA.

Pipeline	Accuracy (%)	Balanced (%)	Macro F1
Our JI-ADF adaptation	53.82	15.72	0.1498
Concatenation control	66.41	28.30	0.2519
Original late fusion	63.55	46.69	0.4026

Original late fusion raises accuracy from 53.82% to 63.55%, balanced accuracy from 15.72% to 46.69%, and macro F1 from 0.1498 to 0.4026 over our adaptation. The concatenation control has higher accuracy but lower macro F1. These complete-pipeline results do not establish superiority to the published model.

Class error analysis: precision, recall and support

The complete class report shows why aggregate improvement needs a diagnostic breakdown. BCC and NV have stronger F1 scores than the remaining categories, while INF and MAL_OTH remain unresolved under the fixed argmax decision. The support column distinguishes a poor estimate based on many observations from one based on a handful of lesions.

Table 4.4. Retained ensemble classification report on validation; zero-division scores are zero.

Class	Precision	Recall	F1	Lesions
AKIEC	0.455	0.444	0.449	45
BCC	0.847	0.820	0.833	378
BEN_OTH	0.667	0.286	0.400	7
BKL	0.432	0.390	0.410	82
DF	0.500	0.500	0.500	8
INF	0.000	0.000	0.000	8
MAL_OTH	0.000	0.000	0.000	1
MEL	0.540	0.507	0.523	67
NV	0.741	0.768	0.754	112
SCCKA	0.515	0.732	0.605	71
VASC	0.444	0.571	0.500	7

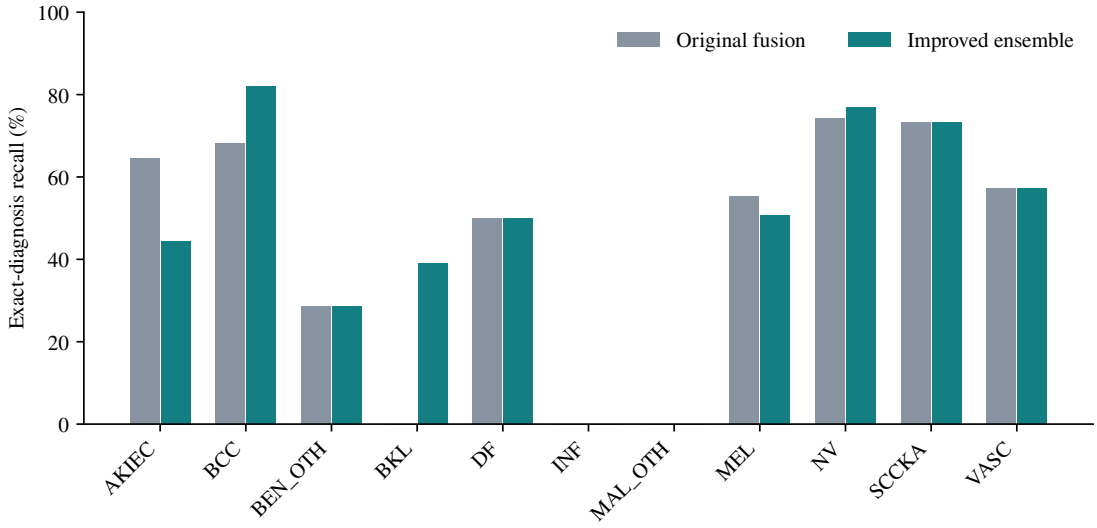


Figure 4.1. Exact-diagnosis recall before and after the frozen extension. Both series use the same validation lesions and class order.

BCC correct predictions increase from 258/378 to 310/378, and BKL from 0/82 to 32/82. Nevus improves from 83/112 to 86/112; SCCKA remains 52/71. AKIEC falls from 29/45 to 20/45 and melanoma from 37/67 to 34/67. Thus the aggregate gain is not uniform across diagnoses. Macro F1 remains a suitable class-balanced summary, but it cannot substitute for these individual recall measurements.

Reading the complete confusion structure

Rows in Figure 4.2 are reference categories and columns are predictions. Values are percentages within each reference class. Row normalization makes the single MAL_OTH case visible alongside the 378 BCC cases; it does not give them equal statistical reliability. Unannotated cells contain zero lesions.

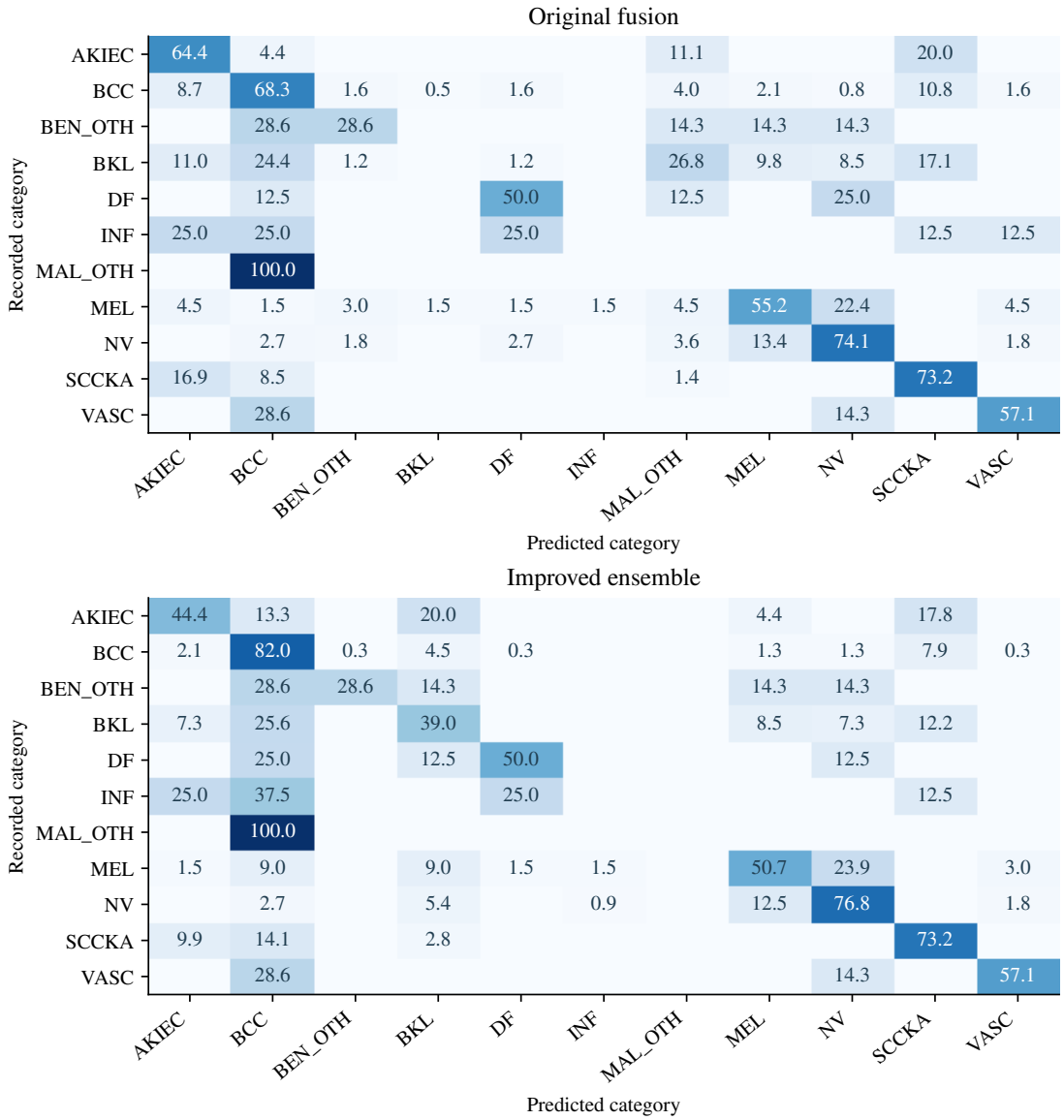


Figure 4.2. Row-normalized validation confusion matrices. Nonzero cells show the percentage of that reference class assigned to each prediction, rounded for display. Counts and class supports remain available in the class tables.

The stronger BCC and BKL diagonals account for much of the increased accuracy. Off-diagonal cells remain important for MEL, AKIEC and the uncommon categories. The matrix identifies where to inspect examples and labels; it does not establish a visual or biological cause for a particular confusion.

Prediction volume and recurring error destinations

The number of predictions assigned to a category helps interpret its precision. The ensemble predicts BCC 366 times, of which 310 are correct. It predicts BKL 74 times, with 32 correct, and MEL 63 times, with 34 correct. These counts distinguish recovery of a class from indiscriminate overprediction of that class.

Table 4.5. Prediction frequency and leading ensemble confusions on the same validation lesions.

Class	True n	Predicted n	Leading wrong destinations
AKIEC	45	44	BKL (9), SCCKA (8)
BCC	378	366	SCCKA (30), BKL (17)
BEN_OTH	7	3	BCC (2), BKL (1)
BKL	82	74	BCC (21), SCCKA (10)
DF	8	8	BCC (2), BKL (1)
INF	8	2	BCC (3), AKIEC (2)
MAL_OTH	1	0	BCC (1)
MEL	67	63	NV (16), BCC (6)
NV	112	116	MEL (14), BKL (6)
SCCKA	71	101	BCC (10), AKIEC (7)
VASC	7	9	BCC (2), NV (1)

Melanoma and nevus

Sixteen of the 67 melanoma lesions are classified as NV; six each are assigned to BCC and BKL. Fourteen of the 112 NV lesions are assigned to MEL. These reciprocal errors justify focused review of the corresponding paired inputs and their reference annotations. Their occurrence alone does not identify whether preprocessing, visual representations, supervision or the downstream boundary is the dominant cause.

NV-to-DF errors fall from three to zero in this validation cohort. That measured change does not imply that all nevus inputs are now recognized: 26 NV cases still receive another category. The originally inspected holdout nevus also remains a separate misclassification, as described in the demonstration section.

The AKIEC precision–recall trade-off

AKIEC recall decreases from 64.44% to 44.44%, while precision increases from 32.95% to 45.45%. Its F1 consequently changes only from about 0.436 to 0.449. This is a concrete example of why an F1 improvement cannot be read as improved recovery of every category. The model makes fewer AKIEC calls, but misses more reference AKIEC lesions. In the uncommon categories, INF has two predictions and no true positives; MAL_OTH receives no leading-category prediction.

Discussion of the measured improvements

The retained extension changes three downstream stages: the information reaching the classifier, the aggregation of transformed image predictions and the combination of the two members. The original image encoders and metadata MLP remain fixed. This permits a practical improvement without repeating their training, while retaining the original pipeline as a directly selectable research comparator.

Frozen embeddings may retain distinctions compressed by the original heads, while view averaging can supply complementary predictions. These are interpretations of the measured result, not independently proven mechanisms.

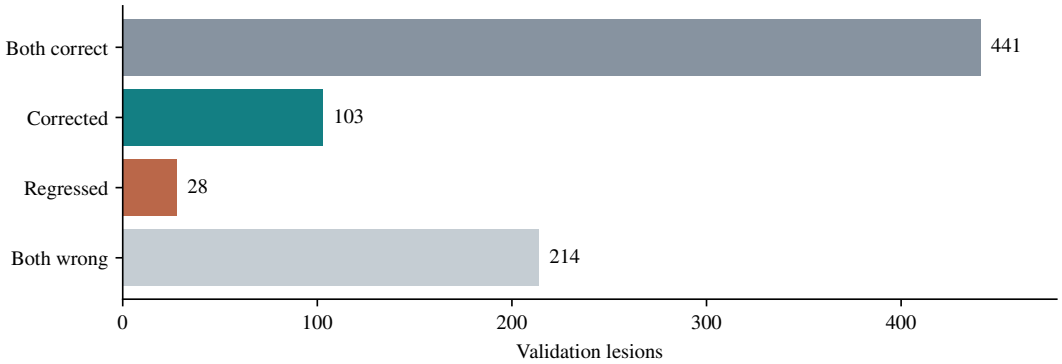


Figure 4.3. Paired correctness transitions for all 786 validation lesions. The extension corrects 103 errors and regresses 28 correct decisions, for a net gain of 75 correct cases.

Conditional uncertainty and attribution

Paired bootstrap intervals resample the same lesions 5,000 times, conditional on the observed predictions. They do not adjust for sequential candidate exploration or training-seed variation.

Table 4.6. Recorded paired improvement estimates and conditional 97.5% intervals.

Metric difference	Estimate	Conditional interval
Accuracy	0.0954	[0.0649, 0.1272]
Balanced accuracy	0.0282	[0.0062, 0.0504]
Macro F1	0.0972	[0.0578, 0.1342]

The preset 0.10 macro-F1 gain target is not met: the observed increase is 0.0972. The recorded target is retained. Component selection was not fully nested; the feature head also uses 3,930 train-plus-fusion labels, compared with 524 fitting labels for original fusion. The measured gain is therefore a complete-pipeline result, rather than a causal estimate for one fusion component.

4.3 Broader Screening: Matched Baseline Comparison

The same 786 validation lesions contain 562 positive-family and 224 negative-family cases. Always predicting the positive family gives 71.50% accuracy. Each table row below uses the same reference grouping, cases and rule for both pipelines. The current demonstration uses grouped leading diagnoses; the summed-score rows preserve the separate archived experiment.

Table 4.7. Broader screening under both fixed decision rules. Sensitivity and specificity refer to the declared families.

Rule	Pipeline	Acc. (%)	Sens. (%)	Spec. (%)	Macro F1
Grouped top	Original	80.41	91.81	51.79	0.7356
Grouped top	Improved	82.70	88.97	66.96	0.7842
Sum ≥ 0.5	Original	40.97	21.00	91.07	0.4025
Sum ≥ 0.5	Improved	83.72	90.04	67.86	0.7957

Grouping the existing top diagnosis increases accuracy from 80.41% to 82.70% (+2.29 points) and specificity from 51.79% to 66.96% (+15.18 points), while sensitivity decreases from 91.81% to 88.97% (-2.85 points). Positive-family misses increase from 46 to 62 and benign/inflammatory false alarms decrease from 108 to 74. This is a trade-off rather than improvement in every screening measure.

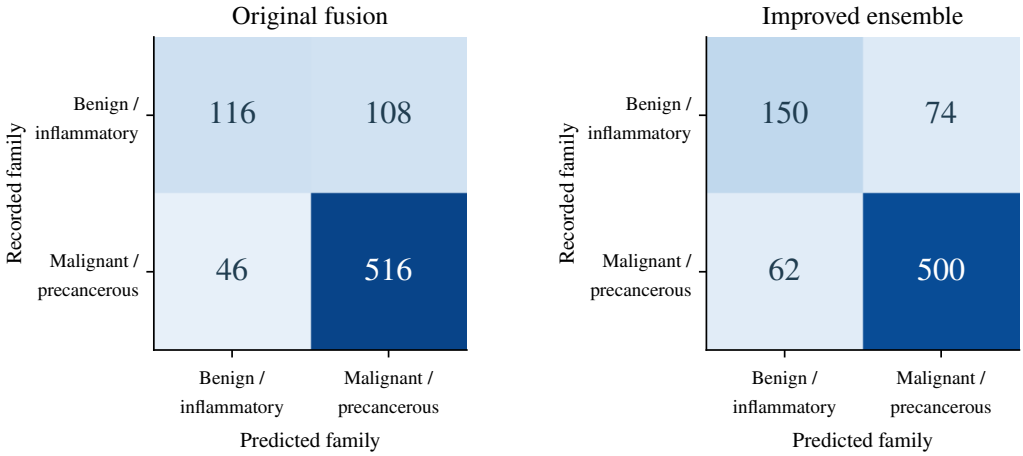


Figure 4.4. Broader screening under the presentation policy: group each model's leading eleven-class diagnosis. Rows are reference families; columns are predicted families.

Table 4.8. Complementary screening measures for the presentation policy on the same 786 lesions.

Pipeline	Balanced (%)	Precision (%)	NPV (%)	Macro F1
Original fusion	71.80	82.69	71.60	0.7356
Improved ensemble	77.97	87.11	70.75	0.7842

Precision summarizes correctness among positive calls; NPV summarizes correctness among negative calls. Both describe this cohort's distribution, not individual risk in a new population.

The archived summed-score experiment

With summed scores at 0.5, the ensemble achieves 83.72% accuracy, 78.95% balanced accuracy, 90.04% sensitivity and 67.86% specificity. Positive precision is 87.54%, positive-class F1 is 0.8877 and negative predictive value is 73.08%. It detects 506/562 positive-family lesions, misses 56 and generates 72 false alarms among 224 negative-family cases. The accuracy Wilson 95% interval is 80.97–86.13%.

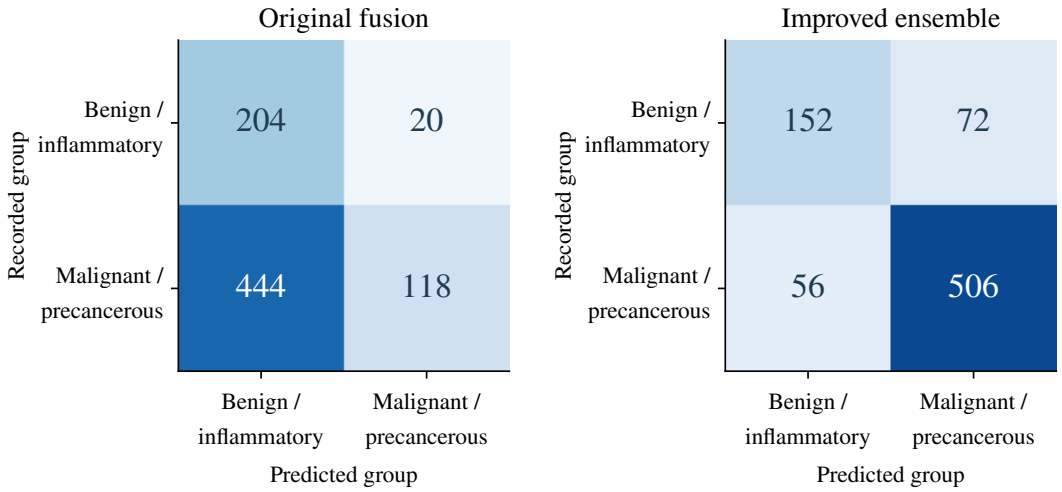


Figure 4.5. Archived broader screening counts with the same summed-score cutoff of 0.5. This is distinct from the demonstration's category-grouping policy.

The original summed-score accuracy is 40.97%, despite 80.41% accuracy when its existing top diagnosis is regrouped. Its grouped score exceeds 0.5 for only 138/786 cases. The large threshold-specific gap reflects how differently the two uncalibrated output distributions interact with that cutoff; it should not be presented as a comparable gain from matched binary-classifier training.

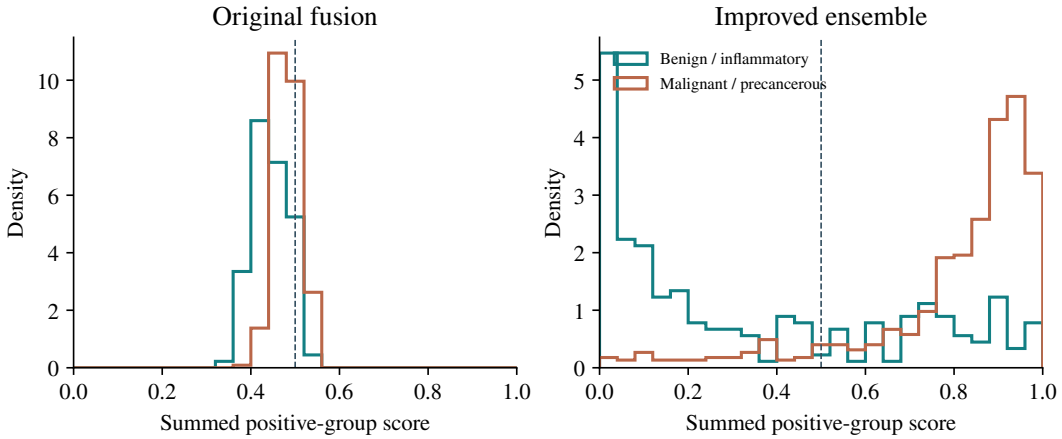


Figure 4.6. Grouped score distributions by reference family. Density is normalized within each family; the dashed line is the fixed 0.5 cutoff.

ROC-AUC increases from 0.7597 to 0.8613 and average precision from 0.8514 to 0.9178. Excluding mixed AKIEC and SCCKA categories leaves 670 lesions (446 positive, 224 negative). Renormalized scores yield ensemble accuracy 81.79%, sensitivity 84.98%, specificity 75.45% and ROC-AUC 0.8654. This different subset is not substituted for the full-cohort result.

4.3.1 Diagnosis-Specific Detection and Mixed Labels

High family sensitivity does not establish reliable detection of every malignant diagnosis. Under either recorded rule, melanoma is flagged positive in 41/67 cases (61.19%); 26 are missed. Only 34 are correctly classified as MEL. Seven further melanoma cases receive another positive-family diagnosis: broader detection improves over exact matching without correcting those diagnostic errors.

Table 4.9. Positive-family detection versus exact diagnosis. Misses refer to the leading-diagnosis grouping rule.

Class	*n*	Exact	Grouped top	Pooled	Missed
AKIEC	45	20	36	39	9
BCC	378	310	353	356	25
MAL_OTH	1	0	1	1	0
MEL	67	34	41	41	26
SCCKA	71	52	69	69	2

Table 4.10. False alarms within the benign/inflammatory reference family, using grouped leading diagnoses.

Class	*n*	False alarms	Correct family	False-alarm rate (%)
BEN_OTH	7	3	4	42.86
BKL	82	44	38	53.66
DF	8	2	6	25.00
INF	8	6	2	75.00
NV	112	17	95	15.18
VASC	7	2	5	28.57

BKL accounts for 44 of the 74 false alarms under category grouping. Nevus contributes 17. These counts make review priorities more specific than a single specificity value. MAL_OTH has one validation case, detected at family level but misclassified as BCC; that support cannot establish a stable class-specific screening estimate.

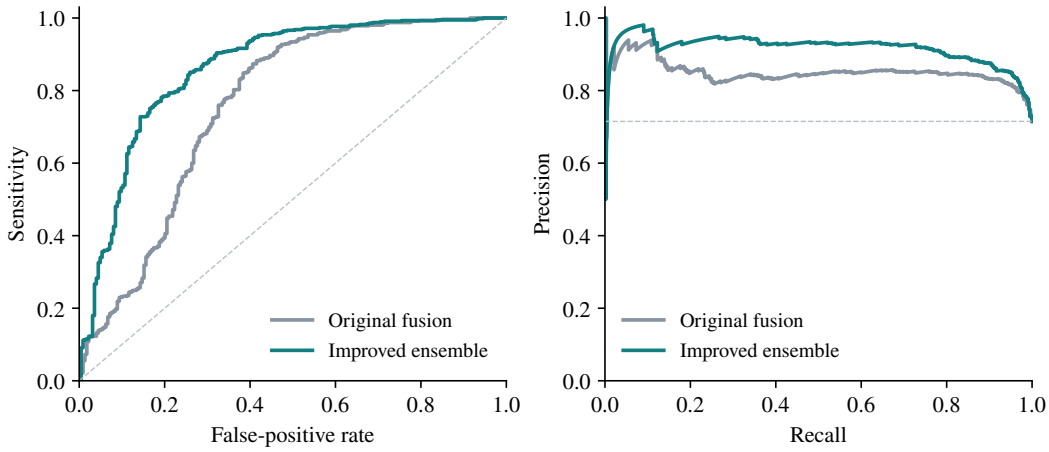


Figure 4.7. Continuous summed-score ROC and precision–recall curves on the same validation lesions. These assess score ranking, independently of the presentation's category-grouping rule.

4.3.2 Historical Comparison with the JI-ADF Adaptation

The earlier adaptation and original late fusion can also be compared on the same 524 archived holdout lesions (376 positive-family, 148 negative-family). Each existing eleven-class prediction is mapped to its family, without score recalibration or model changes. These are our adaptation results, not the published paper's performance, and are separate from the ensemble's validation cohort.

Table 4.11. Historical broader screening on the same 524 holdout lesions; grouped leading diagnosis.

Pipeline	Acc. (%)	Sens. (%)	Spec. (%)	Macro F1
JI-ADF adaptation	77.86	94.15	36.49	0.6707
Concatenation control	81.87	94.68	49.32	0.7440
Original late fusion	82.25	93.62	53.38	0.7564

Original late fusion improves broader accuracy over our JI-ADF adaptation from 77.86% to 82.25% (+4.39 points) and specificity from 36.49% to 53.38%. False alarms fall from 94 to 69, while positive-family misses rise from 22 to 24. This supports improved overall discrimination with a small sensitivity trade-off; it does not establish improved sensitivity for every diagnosis.

Why a common backbone does not settle the comparison

Our JI-ADF adaptation uses paired visual features, a joint attention path, auxiliary image/metadata heads and a sample-dependent gate. The original pipeline trains separate image specialists and fits a later score classifier. Changing these learning relationships changes the complete optimization problem, even when the image-backbone family is shared. The resulting performance difference cannot be assigned to attention or adaptive weighting alone.

Table 4.12. Confirmed configuration distinctions in the completed historical runs.

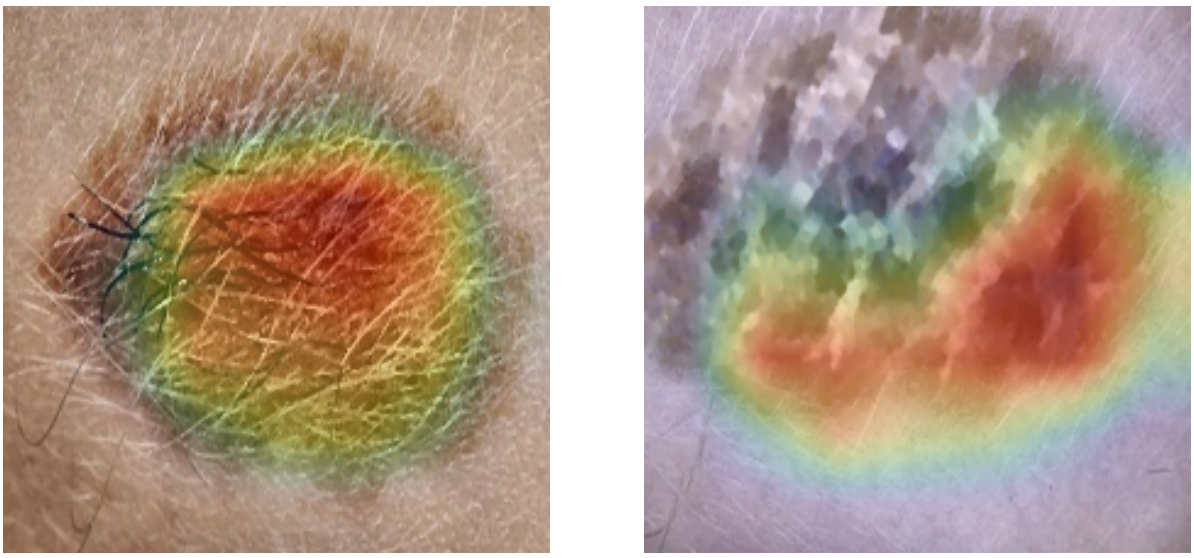
Aspect	Original modular pipeline	Local adaptation / control
Image preparation	224-pixel deterministic evaluation inputs	384-pixel configured input size.
Training organization	Separately trained image and metadata specialists	Coupled image, joint and metadata branches.
Recorded precision	AMP enabled in original image training	Completed adaptation/control use FP32.
Fusion structure	Class-balanced logistic score stack	Adaptive branch decision fusion or fixed concatenation control.

The adaptation record specifies seed 42, batch size four with accumulation over four steps, a maximum of 50 epochs and early-stopping patience ten. The completed FP32 run follows a mixed-precision initialization failure; a successful retry establishes completion, not equivalence to another training recipe. A controlled attribution study would match preprocessing, data budgets, optimization and seeds before changing a single fusion component.

4.4 Demonstration and Verification

The application presents broader screening, the unchanged eleven-class result and model evidence in separate views. Its fixed presentation rule groups the leading diagnosis. The export records that same rule and retains pooled-score analysis separately. An inconclusive diagnostic pattern remains visible; a rejected upload produces neither disease classification nor broader screening.

Nine correctly classified validation pairs, one for each class with a correct case, are explicitly labelled as selected success examples. The nevus `IL_0008891` and melanoma `IL_0499462` examples match their recorded diagnoses. They illustrate execution, not an unbiased success rate. Dataset diagnosis annotations are displayed after inference and never enter the predictor's metadata features.



(a) Clinical specialist

(b) Dermoscopic specialist

Figure 4.8. Grad-CAM overlays for the selected correctly classified nevus. Each map explains that image specialist's leading category, rather than the entire ensemble or screening decision.

The retained original case `IL_0024228` remains wrong at exact diagnosis: recorded NV, original DF and ensemble INF. Both predicted categories nevertheless belong to the reference benign/inflammatory family. This example illustrates why a correct broad-family result must not be presented as a correct eleven-class diagnosis.

Numerical replay matches all 786 archived validation decisions. The raw-image integration check reproduces archived ensemble scores within 3.4×10^{-8}. A separate local CPU package check on the selected nevus reproduces its saved scores exactly and includes both heatmaps in 2.512 seconds, with peak process RSS 880.2 MiB. This is one cold-load macOS measurement, not an average latency or a cloud-platform benchmark.

Forty targeted checks cover score ordering, screening policy, explanation behavior, input rejection and interface consistency. The validator's separate archive reports acceptance of 176/176 supported test images and rejection of 289/289 unrelated images, plus 44/44 bottle stress images [16]. These finite gate results are not disease accuracy or proof of rejection for every possible unrelated photograph.

4.5 Interpretation and Limits

The project demonstrates aggregate gains, class-level insights and consistent presentation of its two endpoints. The following limits concern generalization, attribution and the next experiments.

Rare-class uncertainty

Equal weighting in macro F1 prevents common categories from dominating, but it also makes estimates for tiny classes consequential. Wilson intervals below describe uncertainty conditional on the observed cases. They assume an independent binomial observation model; patient-level independence has not been established, and the intervals are not simultaneous guarantees across all categories.

Table 4.13. Selected exact-diagnosis recalls with descriptive Wilson 95% intervals.

Class	Correct / n	Recall (%)	95% interval (%)
BCC	310/378	82.01	77.82–85.55
INF	0/8	0.00	0.00–32.44
MAL_OTH	0/1	0.00	0.00–79.35
MEL	34/67	50.75	39.06–62.35
NV	86/112	76.79	68.16–83.64

Zero correct MAL_OTH predictions among one case cannot establish a precisely estimated population recall of zero. Conversely, detecting its sole case at broader family level cannot establish reliable screening for that category. The larger melanoma denominator supports a more informative estimate, but its interval remains wide enough to motivate additional independent cases.

Development-set reuse and causal attribution

The validation set has been repeatedly inspected, including original checkpoint selection and sequential downstream studies. Final fusion OOF selection is not fully nested. Methods also have different downstream label budgets, and only one original training seed is available. Conditional paired intervals do not resolve these design limits. Future comparisons should lock the full selection procedure and evaluate matched models on a new patient-grouped cohort.

Coverage, calibration and explanation

Skin-tone and rare-class coverage are uneven; six complete-release lesions in the least represented skin-tone category cannot support strong subgroup performance claims. Acquisition differences and heuristic crops deserve systematic review. Scores remain uncalibrated, and Grad-CAM is a branch-level explanation rather than a correctness test. The relevance gate addresses input suitability, not every form of distribution shift.

Priority of further work

The next experiments should first verify labels, paired-image quality and recurring confusion groups, then compare matched-budget downstream learners. Selective encoder fine-tuning can subsequently target persistent errors in a separate controlled run. Calibration and any sensitivity-driven operating-point selection should use separate fitting data. These are prospective steps; this revision fits no model and changes no saved prediction.

CHAPTER 5

CONCLUSION AND FUTURE WORK

The frozen-feature/TTA extension improves eleven-class validation macro F1 from 0.3551 to 0.4523 while retaining the original image specialists and metadata MLP. Exact-class accuracy increases from 59.67% to 69.21%, with 103 previous errors corrected and 28 correct decisions regressed. These findings demonstrate useful aggregate progress without requiring another image-backbone training run. The persistent melanoma and rare-class errors also show why the improvement should be characterized through both aggregate and category-specific evidence.

Broader screening provides a complementary endpoint. Grouping the leading diagnosis raises matched validation accuracy from 80.41% to 82.70%, with improved specificity and reduced sensitivity. This is the single presentation policy used by the local demo. The archived summed-score experiment reaches 83.72% accuracy and ROC-AUC 0.8613; its fixed cutoff remains uncalibrated and is reported separately. Screening accuracy and exact diagnostic accuracy answer different questions and are not interchangeable measures of success.

The earlier JI-ADF adaptation comparison remains a historical reference. The main improvement claim uses the matched original-to-ensemble validation comparison; it does not substitute the published paper's benchmark or the historical holdout for that cohort.

Table 5.1. Contributions and the evidence supporting them.

Contribution	Evidence retained in this report
Practical multimodal extension	Frozen-feature and view-averaging members; complete matched validation comparison.
Transparent error analysis	Class support, precision/recall, confusion destinations and paired correctness changes.
Consistent screening endpoint	Declared diagnostic families, fixed rules, matched counts and class-specific detection.
Reproducible demonstration	Structured outputs, input rejection, branch heatmaps, numerical replay and preserved baseline artifacts.

Further work should prioritize independent patient-grouped evaluation and additional examples for the underrepresented categories. Matched-label-budget studies and multiple training seeds would clarify how much of the gain is attributable to representation access, fitting data or prediction averaging. Targeted image-quality review and selective fine-tuning can then address the observed confusion groups. Calibration should precede any interpretation of scores as individual risk or any sensitivity-driven threshold selection.

The project provides an evaluated three-modality EfficientNetV2 ensemble, a reproducible interpretation pipeline and a measured improvement over its saved original validation reference. Persistent errors, including melanoma family sensitivity of 61.19%, define concrete priorities for further study. Trained weights and historical results remain preserved.

APPENDIX A

SUPPLEMENTARY BINARY CLASSIFICATION STUDY

A.1 Purpose and Experimental Separation

This supplementary experiment asks whether direct binary supervision and availability-aware fusion can improve malignancy screening when an image modality is missing. It is separate from the main eleven-class EfficientNetV2 ensemble and was not installed into the demonstration. The completed Kaggle Version 4 ran for 4,764.3 seconds on two Tesla T4 GPUs [17].

The binary positive endpoint includes source-labelled BCC, melanoma, invasive SCC, SCC in situ and malignant collision lesions. Actinic keratosis and keratoacanthoma are excluded; the remaining eligible nonmalignant diagnoses form the negative class. This differs from the main report's broader diagnostic-family grouping, which retains mixed AKIEC and SCCKA categories. Consequently, the appendix's accuracy, macro F1 and positive-class Dice cannot be substituted for the main results.

Table A.1. Cases retained after endpoint and overlap checks in the supplementary study.

Source	Image inputs	Train	Val.	Cal.	Test
MILK10k	Clinical + dermoscopic	3205	733	491	484
ISIC 2019	Dermoscopic	9150	1975	1958	—
PAD-UFES-20	Clinical	—	—	—	1236

MILK partitions retain their original assignments. ISIC 2019 provides additional development dermoscopy [18]; PAD-UFES-20 is a separately sourced clinical-image test [19]. Missing modalities remain absent. Shared predictors use age, sex, coarse anatomical site and observation flags. Images with exact, identity or conservative perceptual overlap are quarantined by group. Patient grouping is used when supplied, but is unavailable for the MILK and ISIC development records. The retained PAD cohort contains 1,236 lesions from 960 patients: 833 positive and 403 negative cases.

Five methods receive the same frozen encoders and permitted data: B0 logistic regression, B1 fixed-fusion MLP, B2 fixed fusion with subtype supervision, B3 reliability fusion and B4 reliability fusion with subtype supervision. Neural heads average three training seeds. PCA with 128 coordinates per image branch and metadata transforms fit training data only. Checkpoint selection uses validation data; calibration and sensitivity-targeting thresholds use the calibration split. Model and threshold identities are locked before PAD access. The 0.95 sensitivity setting is a calibration target, not a promised external sensitivity.

A.2 External Screening Results

Table A.2 reports clean clinical images plus metadata at independently locked thresholds. All methods use the same 1,236 PAD lesions. The images are classified as positive or negative; these scores do not measure exact recognition among eleven diagnostic categories.

Table A.2. PAD external results for clinical image plus metadata.

Method	Acc. (%)	ROC-AUC	Sens. (%)	Spec. (%)	Macro F1
B0 Logistic	77.43	0.8654	98.92	33.00	0.6716
B1 Fixed fusion	78.88	0.9313	98.92	37.47	0.6998
B2 Fixed + subtype	79.45	0.9322	99.04	38.96	0.7097
B3 Reliability	79.37	0.9266	99.04	38.71	0.7082
B4 Reliability + subtype	79.61	0.9325	98.56	40.45	0.7155

B4 has accuracy 79.61%, ROC-AUC 0.9325, balanced accuracy 69.50%, macro F1 0.7155 and positive-class F1/Dice 0.8669. It detects 821/833 positive lesions and correctly excludes 163/403 negative lesions. This yields 12 false negatives and 240 false positives. Its high sensitivity therefore coexists with a substantial benign false-alarm burden.

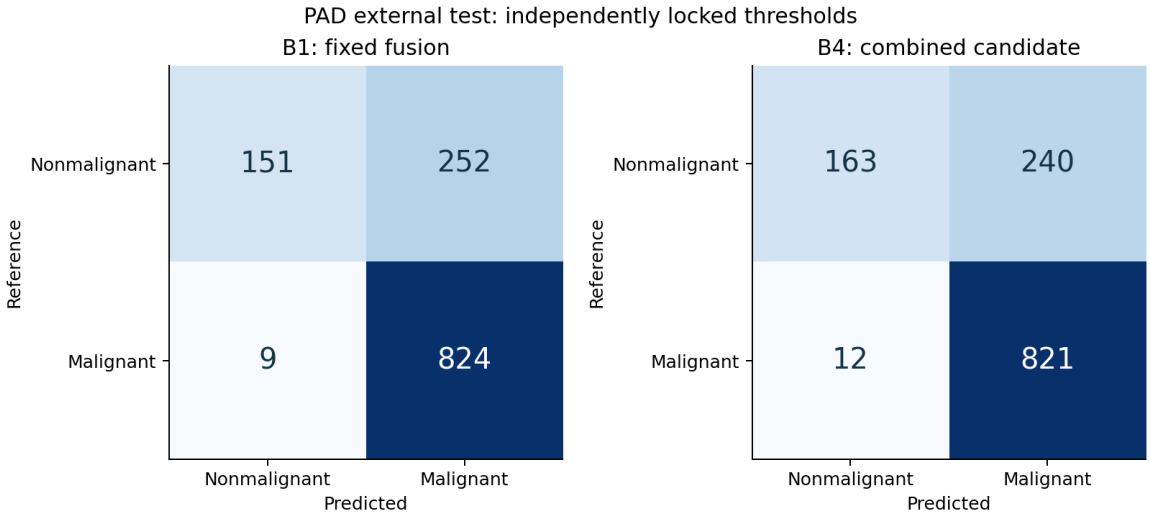


Figure A.1. External error counts for the predeclared fixed-fusion comparator B1 and combined candidate B4. Each uses its own threshold locked on calibration data.

B4 reduces false alarms by 12 relative to B1 while adding three malignant misses. It detects 33/36 melanoma lesions (91.67%), compared with 34/36 for B1. The melanoma denominator is small; B4's Wilson 95% interval is 78.17–97.13%. Overall malignant sensitivity is dominated by the much more frequent BCC cases and must be read alongside subtype results.

A.3 Uncertainty and Relationship to the Main Ensemble

The predeclared primary comparison is B4 minus B1. A paired bootstrap resamples 960 PAD patient groups 1,000 times. Table A.3 reports percentage-point differences at the locked operating points; it is not a comparison at an identical realized external sensitivity.

Table A.3. External paired patient-bootstrap comparison, B4 minus B1.

Measure	Difference (pp)	95% interval (pp)
Specificity	+2.98	[+0.70, +5.46]
Malignant sensitivity	-0.36	[-0.82, +0.00]
Melanoma sensitivity	-2.78	[-9.68, +0.00]
Macro F1	+1.56	[+0.14, +3.06]

The specificity interval supports a modest reduction in benign false alarms. The planned overall criterion was not met: the melanoma sensitivity interval fails the preset preservation limit of two percentage points. The proposed combination's overall advantage remains unresolved.

Table A.4. Supplementary paired MILK holdout: the same 484 eligible lesions.

Method	Acc. (%)	Sens. (%)	Spec. (%)	Macro F1
Archived original grouping	80.99	92.98	52.11	0.7451
B0 Logistic	82.02	95.61	49.30	0.7497
B1 Fixed fusion	81.82	95.91	47.89	0.7444
B4 Reliability + subtype	80.99	95.91	45.07	0.7294

The combined candidate does not improve on fixed fusion on this internal holdout. The archived original comparator is the original eleven-class argmax mapped to the new eligible binary endpoint; it is not the later improved feature/TTA ensemble. The main report's 82.70% broader-screening result concerns different cases, labels and a different decision rule. Directly ranking those percentages would confound the endpoint and cohort.

The appendix strengthens the experimental record through direct binary controls, separate external-source evaluation, missing-input tests and explicit uncertainty. It does not replace the main ensemble or establish a substantial improvement over it. All 220 archived cohort/method/condition metric groups were independently replayed from saved predictions, and 36 locked artifact identities were verified. Baseline checkpoints, the active demonstration and the original report evidence remain unchanged. Further encoder tuning requires a separate experiment; PAD results are now inspected and cannot guide tuning while remaining an uninspected confirmation set.

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